

TINProofs C1-C5

Formal Lemmas and Theorems

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C1. Commodity Hull Theorem

NOTATION. For a feasible path $p = (T_p, Q_p)$, define

$$A_p(\lambda) = -\log Q_p + \lambda T_p, \quad V_p(\lambda) = \log Q_p - \lambda T_p.$$

For a finite nonempty hull C ,

$$F_C(\lambda) = \inf_{p \in C} A_p(\lambda), \quad V_C(\lambda) = \sup_{p \in C} V_p(\lambda).$$

Affine path action and value

action_affine; value_affine; action_neg_value

For $a, b \geq 0$ with $a + b = 1$,

$$A_p(a\lambda_1 + b\lambda_2) = aA_p(\lambda_1) + bA_p(\lambda_2), \quad V_p(a\lambda_1 + b\lambda_2) = aV_p(\lambda_1) + bV_p(\lambda_2),$$

and $A_p(\lambda) = -V_p(\lambda)$.

Convexity and concavity corollaries: action_convex0n; action_concave0n;
value_convex0n; value_concave0n.

Finite hull support and value duality

support_concave0n; valueFn_eq_neg_support; valueFn_convex0n;
legendre_duality

The finite lower envelope F_C is concave on univ, the finite upper envelope V_C is convex on univ, and

$$V_C(\lambda) = -F_C(\lambda).$$

This is the formal Legendre-duality surface for the finite path hull.

Crossover equality

action_eq_at_crossover

For distinct exposures $T_i \neq T_j$, set

$$\lambda_{ij} = \frac{\log Q_j - \log Q_i}{T_j - T_i}.$$

Then the path actions agree exactly:

$$A_i(\lambda_{ij}) = A_j(\lambda_{ij}).$$

Crossover ordering

`action_lt_below_crossover; action_gt_above_crossover`

If $T_i < T_j$, then

$$\lambda < \lambda_{ij} \Rightarrow A_j(\lambda) < A_i(\lambda), \quad \lambda_{ij} < \lambda \Rightarrow A_i(\lambda) < A_j(\lambda).$$

Thus the theorem identifies both the crossing point and the side on which each path is selected.

C2. Stochastic Certification Asymmetry

NOTATION. Let $k > 0$ be the reserve-cycle count, let tier thresholds be $\tau_i \in (0, 1)$, and define the integer cutoff

$$x_i = \lceil k\tau_i \rceil.$$

For $X \sim \text{Binom}(n, p)$,

$$B_{n,p}(j) = \binom{n}{j} p^j (1-p)^{n-j}.$$

Absorbing lowest-tier cutoff

`ceil_mean_lt_xMin_of_lt_sub_one; median_lt_xMin_of_median_le_ceil_mean;`
`absorbing_lowest_tier_median_lt_cutoff`

If $p \geq 0$ and

$$np < k\tau_i - 1,$$

then

$$\lceil np \rceil < x_i.$$

Consequently, any median m with $m \leq \lceil np \rceil$ lies below the same cutoff. The lowest-tier theorem is this statement specialized to the final threshold $i = i_{\min}$.

The mode consequence is recorded separately as `mode_lt_xMin_of_mode_le_median`.

Tier asymmetry as finite sums

`TierSystem.xMin_pos; binomialCDF_pred_eq_sum_range;`
`tierAsymmetryRatio_eq_binomial_sums`

The tier-specific downgrade-to-upgrade asymmetry ratio is exactly

$$R_{T,T^-} = \frac{\sum_{j=0}^{x_T-1} B_{n,p}(j)}{\sum_{j=x_T-}^n B_{n,p}(j)}.$$

The proof first shows each cutoff x_i is positive, so the lower tail is the range $0, \dots, x_T - 1$.

Binomial-to-Poisson tail limit

`binomialPMF_tendsto_poissonPMF; binomialLowerTail_tendsto_poissonLowerTail;`
`binomialUpperTail_tendsto_poissonUpperTail`

If $np_n \rightarrow r$, then for each fixed k ,

$$B_{n,p_n}(k) \longrightarrow e^{-r} \frac{r^k}{k!}.$$

The corresponding finite lower tails and upper tails converge to the Poisson

tails

$$L_m(\mu) = \sum_{j=0}^m e^{-\mu} \frac{\mu^j}{j!}, \quad U_k(\mu) = \sum_{j \geq k} e^{-\mu} \frac{\mu^j}{j!}.$$

Supporting normalizations and complement identities:

`binomialPMF_sum_range_eq_one; poissonUpperTail_eq_one_sub_lower;`
`hasSum_poissonPMF.`

Poisson crossover existence

`poissonCrossoverCondition_iff; poissonCrossover_exists_of_bracket;`
`poissonCrossover_exists_of_reverse_at`

The limiting crossover equation is

$$L_{k-2}(\mu) = U_k(\mu).$$

If $a \leq b$, $U_k(a) \leq L_{k-2}(a)$, and $L_{k-2}(b) \leq U_k(b)$, then there exists

$$\mu \in [a, b] \quad \text{with} \quad L_{k-2}(\mu) = U_k(\mu).$$

The specialized theorem uses $a = 0$ and any nonnegative right endpoint where the inequality has reversed.

Asymptotic delivery-ratio expansion

`adellJodraAsymptoticInput; drStar_expansion_of_lambda_expansion`

Under the named asymptotic input

$$\lambda_k^* = k - \frac{5}{6} + \mathcal{O}(k^{-1}),$$

the selected delivery ratio

$$\text{DR}_k^* = \frac{\lambda_k^*}{k}$$

satisfies

$$\text{DR}_k^* = 1 - \frac{5}{6k} + \mathcal{O}(k^{-2}).$$

This theorem is conditional on the external Poisson-median asymptotic boundary.

Gap tier

`no_achievable_in_gap; gap_tier`

If two adjacent ceiling cutoffs coincide,

$$x_i = x_j,$$

then no integer X satisfies

$$x_i \leq X < x_j.$$

C3. Lyapunov Radius of Validity

NOTATION. The setup stores positive constants $\alpha_{\min}, \alpha_{\max}, q, r_o, \|A\|, M$ and defines

$$r_* = \min \left(r_o, \frac{q}{2\|A\|M} \right), \quad c_* = \frac{q}{2\alpha_{\max}}.$$

For a perturbation point P , write $\rho = \|\delta\|$, $V = P.V$, and $\dot{V} = P.V \dot{}$.

Eigenvalue sandwich

`eigen_sandwich_lower`; `eigen_sandwich_upper`; `eigen_sandwich`

The Lyapunov quadratic value satisfies

$$\alpha_{\min}\rho^2 \leq V \leq \alpha_{\max}\rho^2.$$

The lower and upper inequalities are available both separately and as a conjunction.

Derivative decomposition and pointwise bounds

`vdot_decomposition`; `quadratic_term_lower`; `negative_quadratic_term_upper`;
`remainder_abs_bound`; `remainder_upper_bound`

The derivative decomposes as

$$\dot{V} = -Q(\delta) + R(\delta),$$

with

$$q\rho^2 \leq Q(\delta), \quad |R(\delta)| \leq \|A\|M\rho^3.$$

Hence $-Q(\delta) \leq -q\rho^2$ and $R(\delta) \leq \|A\|M\rho^3$.

Critical-ball derivative bound

`vdot_le_quadratic_plus_cubic`; `vdot_bound_on_ball`

Combining the previous inequalities gives

$$\dot{V} \leq -q\rho^2 + \|A\|M\rho^3.$$

Inside the critical ball $\rho < r_*$, the cubic term consumes at most half the quadratic margin:

$$\dot{V} \leq -\frac{q}{2}\rho^2.$$

Main radius theorem

`criticalRadius_pos`; `lyapunov_radius_of_validity`

The radius r_* is positive. If $\rho < r_*$, then

$$\dot{V} \leq -c_* V = -\frac{q}{2\alpha_{\max}} V.$$

This is the formal Lyapunov radius-of-validity theorem.

Sublevel-set corollaries

sublevel_inside_critical_ball; sublevel_boundary_vdot_nonpos

If $V \leq V_o$ and

$$V_o < \alpha_{\min} r_*^2,$$

then $\rho < r_*$. On the boundary $V = V_o$ of such a sublevel set,

$$\dot{V} \leq 0.$$

C4. Temporal Transport Factorization

NOTATION. Let $D \subseteq F$ be delivery and feasibility events in a probability space. Define

$$\text{DR} = \mathbb{P}(D), \quad S_{\text{T}} = \mathbb{P}(F), \quad \eta = \mathbb{P}(D)/\mathbb{P}(F).$$

Delivery implies feasibility

`delivery_inter_infeasible_eq_empty; delivery_zero_when_infeasible;`
`delivery_measure_le_feasible`

The event-level inclusion $D \subseteq F$ yields

$$D \cap F^c = \emptyset, \quad \mathbb{P}(D \cap F^c) = 0, \quad \mathbb{P}(D) \leq \mathbb{P}(F).$$

Exact factorization

`exact_factorization; exact_factorization_real`

Whenever $S_{\text{T}} \neq 0$,

$$\text{DR} = S_{\text{T}} \eta.$$

The theorem is proved both in the extended nonnegative-real measure layer and in the real-valued ratio layer.

Augmentation monotonicity

`st_monotone; stReal_monotone`

For an augmentation $F \subseteq F'$ with the same underlying measure,

$$S_{\text{T}} \leq S'_{\text{T}}.$$

The same monotonicity holds after conversion to real-valued factors.

Braess localization

`braess_in_eta; braess_in_etaReal`

If an augmentation has $S'_{\text{T}} > 0$, preserves the measure, and nevertheless decreases delivery ratio,

$$\text{DR}' < \text{DR},$$

then the decrease is localized in the efficiency factor:

$$\eta' < \eta.$$

C5. Three-Factor Sparse Law

NOTATION. The scalar setup records S_T , η , $E[H]$, a Lyapunov exponent $\lambda < 0$, and $DR = S_T \eta$. Define

$$\eta_{\text{lyap}} = \exp(E[H]\lambda), \quad \Phi = \eta / \eta_{\text{lyap}}.$$

Exact three-factor law

`three_factor`

The sparse-law delivery ratio factors as

$$DR = S_T \eta_{\text{lyap}} \Phi.$$

This is an algebraic refinement of $DR = S_T \eta$ by inserting the chain attenuation factor and residual routing distortion.

Chain attenuation properties

`etaLyap_pos`; `etaLyap_lt_one`; `phi_pos`

The chain attenuation is strictly positive,

$$0 < \eta_{\text{lyap}},$$

and, when $E[H] > 0$ and $\lambda < 0$,

$$\eta_{\text{lyap}} < 1.$$

If $\eta > 0$, then $\Phi > 0$.

Negative mean log-hop exponent

`lyapunov_exponent_neg`

For finitely many hop probabilities $p_i \in (0, 1)$ with $n > 0$,

$$\frac{1}{n} \sum_{i=1}^n \log p_i < 0.$$

This supplies the sign condition used by the chain attenuation factor.

Morphology vocabulary

`morphologySlope`; `isTrap`; `isCluster`

For two residual measurements (Φ_1, E_1) and (Φ_2, E_2) with positive residuals and $E_1 \neq E_2$, define

$$\gamma = \frac{\log \Phi_2 - \log \Phi_1}{E_2 - E_1}.$$

The formal classes are $\gamma < 0$ for a trap and $\gamma > 0$ for a cluster.